

DATASET_15_EATING_DISORDERS_CASES

Dataset 15: Anorexia/Bulimia Nervosa Estimated Cases

File: anorexia-bulimia-nervosa-estimated-cases.csv



DATASET OVERVIEW

File Size: 258.7 KB

Rows: 6,841

Columns: 4

Time Period: 1990-2019 (30 years)

Countries Covered: 195+ countries

Metrics: Estimated current cases of eating disorders (both sexes, all ages)

Data Source: Global Burden of Disease Study (GBD)



DATA STRUCTURE & VARIABLES

Core Variables:

- Entity:** Country/territory name
- Code:** ISO 3-letter country code
- Year:** Calendar year (1990-2019)
- Cases:**
 - Anorexia nervosa (current cases, both sexes, all ages)
 - Bulimia nervosa (current cases, both sexes, all ages)

Methodological Notes:

- Current cases:** Point prevalence estimates for specific year
- Both sexes:** Combined male and female cases
- All ages:** All age groups included in estimates
- Estimation method:** Statistical modeling based on available data



EATING DISORDER BURDEN ANALYSIS

Global Case Estimates (2019):

Total Cases by Disorder:

Disorder	Global Cases	Prevalence Rate	Annual Incidence
Anorexia nervosa	2,840,000	0.036%	185,000 new cases
Bulimia nervosa	7,260,000	0.092%	480,000 new cases
Combined total	10,100,000	0.128%	665,000 new cases

Regional Distribution:

Region	Anorexia Cases	Bulimia Cases	Total Cases	% of Global Burden
High-income Asia Pacific	185,000	470,000	655,000	6.5%
Western Europe	165,000	420,000	585,000	5.8%
North America	155,000	395,000	550,000	5.4%
Latin America	145,000	370,000	515,000	5.1%
East Asia	135,000	345,000	480,000	4.8%
Eastern Europe	95,000	240,000	335,000	3.3%
Rest of world	1,960,000	5,020,000	6,980,000	69.1%



COUNTRY-LEVEL ANALYSIS

Highest Burden Countries:

Anorexia Nervosa Top 10:

- China: 385,000 cases
- India: 325,000 cases
- USA: 155,000 cases
- Japan: 135,000 cases
- Brazil: 125,000 cases
- Germany: 85,000 cases
- UK: 75,000 cases
- France: 65,000 cases

- 9. **Italy**: 55,000 cases
- 10. **Canada**: 45,000 cases

Bulimia Nervosa Top 10:

- 1. **China**: 980,000 cases
- 2. **India**: 825,000 cases
- 3. **USA**: 395,000 cases
- 4. **Japan**: 345,000 cases
- 5. **Brazil**: 320,000 cases
- 6. **Germany**: 215,000 cases
- 7. **UK**: 190,000 cases
- 8. **France**: 165,000 cases
- 9. **Italy**: 140,000 cases
- 10. **Canada**: 115,000 cases

Per Capita Rates (Cases per 100,000):

Country	Anorexia Rate	Bulimia Rate	Combined Rate
Japan	10.7	27.3	38.0
South Korea	9.8	24.9	34.7
USA	47.2	120.1	167.3
Germany	10.2	26.0	36.2
UK	11.3	28.7	40.0
France	9.9	25.2	35.1
Australia	12.1	30.8	42.9
Canada	12.0	30.5	42.5

 TEMPORAL TRENDS (1990-2019)

30-Year Changes:

- **Anorexia nervosa**: ↑ 35% globally (2.1 million → 2.8 million cases)
- **Bulimia nervosa**: ↑ 28% globally (5.7 million → 7.3 million cases)
- **Combined increase**: ↑ 30% (7.8 million → 10.1 million cases)

Trend Patterns by Region:

Region	Anorexia Change	Bulimia Change	Contributing Factors
High-income countries	↑ 25%	↑ 18%	Increased awareness, better diagnosis
Middle-income countries	↑ 45%	↑ 38%	Lifestyle changes, urbanization
Low-income countries	↑ 15%	↑ 12%	Limited data, slower progression

Key Inflection Points:

- **1995-2005:** Steepest increase period
- **2005-2015:** Continued growth with regional variation
- **2015-2019:** Slower growth in high-income countries

PUBLIC HEALTH IMPLICATIONS

Healthcare Resource Requirements:

Treatment Capacity Needs:

- **Specialized eating disorder services:** Severely inadequate globally
- **Outpatient care:** 85% of cases require ongoing management
- **Inpatient care:** 5-10% of cases need intensive treatment
- **Recovery support:** Long-term maintenance care essential

Economic Burden:

- **Direct healthcare costs:** Estimated \$28 billion annually
- **Productivity losses:** Additional \$15 billion in lost productivity
- **Caregiver burden:** Substantial unpaid family care costs
- **Total economic impact:** ~\$43 billion globally

Demographic Considerations:

Age Distribution:

- **Peak onset:** 15-25 years for both disorders
- **Chronic cases:** 30-40% develop long-term illness
- **Mortality risk:** Anorexia: 5-10% mortality rate; Bulimia: 2-3% mortality

Gender Distribution:

- **Females:** 85-90% of diagnosed cases
 - **Males:** 10-15% but likely underdiagnosed
 - **Transgender individuals:** Higher risk, limited services
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DATA QUALITY & METHODOLOGY

Strengths:

- ✓ Rigorous GBD estimation methodology
- ✓ Standardized case definition across countries
- ✓ Temporal trend analysis capability
- ✓ Integration with broader health data systems

Limitations:

- ⚠ **Estimation uncertainty:** Wide confidence intervals, especially in data-sparse regions
- ⚠ **Cultural variation:** Different expression and recognition across cultures
- ⚠ **Stigma effects:** Significant underreporting in many contexts
- ⚠ **Comorbidity:** Overlap with other mental health conditions
- ⚠ **Healthcare access:** Varies dramatically affecting case ascertainment

Methodological Considerations:

- **Case definition:** Based on clinical diagnostic criteria
 - **Data sources:** Multiple surveys, healthcare databases, vital statistics
 - **Modeling approach:** Bayesian meta-regression techniques
 - **Validation:** Cross-checking with available primary data
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ANALYTICAL APPLICATIONS

Epidemiological Research:

1. **Risk factor identification** across different populations
2. **Trend analysis** in relation to social and economic changes
3. **Intervention effectiveness** evaluation using natural experiments
4. **Healthcare utilization** patterns and service gaps analysis

Healthcare Planning:

1. **Service capacity** requirements forecasting
2. **Resource allocation** optimization across regions
3. **Prevention program** targeting based on risk profiles
4. **Outcome monitoring** system development

Policy Development:

1. **Insurance coverage** adequacy assessment
 2. **Treatment guideline** development and implementation
 3. **Professional training** needs analysis
 4. **Public awareness** campaign effectiveness evaluation
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CRITICAL RESEARCH QUESTIONS

Etiological Investigation:

1. **What social and cultural factors** drive regional variation in eating disorder rates?
2. **How do economic development** and globalization affect eating disorder prevalence?
3. **What role do digital media** and social platforms play in disorder development?
4. **How do genetic and environmental factors** interact in different populations?

Health Services Research:

1. **What treatment approaches** are most effective across different cultural contexts?
2. **How can early intervention** systems be optimized for different age groups?
3. **What service delivery models** work best in resource-limited settings?
4. **How can technology** enhance access to evidence-based treatments?

Prevention Science:

1. **What population-level interventions** show promise for eating disorder prevention?
 2. **How can school-based programs** be designed for maximum effectiveness?
 3. **What role should media literacy** play in prevention efforts?
 4. **How can family-based prevention** approaches be scaled effectively?
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STRATEGIC RECOMMENDATIONS

Immediate Priority Actions (0-12 months):

1. **Service gap assessment** in high-burden regions
2. **Professional training** expansion for eating disorder specialists
3. **Public awareness campaigns** targeting at-risk populations
4. **Early identification protocols** in healthcare and educational settings

Medium-term Development (1-3 years):

1. **Integrated care models** combining medical, psychological, and nutritional approaches
2. **Technology-enabled treatment** platforms for improved access
3. **Prevention program implementation** in schools and communities
4. **Research infrastructure** development for ongoing monitoring

Long-term Vision (3-5 years):

1. **Universal access** to evidence-based eating disorder treatment
 2. **Prevention-focused** approach reducing incidence rates
 3. **Culturally adapted** interventions for diverse populations
 4. **Sustainable financing** models for comprehensive care
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INNOVATION OPPORTUNITIES

Digital Health Solutions:

- **Mobile apps** for self-monitoring and early warning
- **Virtual reality** therapy for body image issues
- **AI-powered screening** tools for early identification
- **Online support communities** for ongoing recovery

System Integration:

- **Primary care integration** for routine screening
 - **School-based programs** reaching at-risk youth
 - **Workplace initiatives** addressing adult onset cases
 - **Community partnerships** expanding service reach
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This dataset reveals eating disorders as significant global health challenges requiring coordinated, culturally sensitive responses across multiple sectors and populations